

PEDRO HENRIQUE MASTEGUIN

Mid-Level Developer at Segura | Electronic Technician | Embedded Systems, Software Development, Computer Vision,
Fullstack, Cybersecurity | C, C++17, Bash, Python, Lua, Vue3, Linux, Git, Docker

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Education

Bachelor of Science and Technology Mar 2021 – Dec 2023

UNIFESP - Universidade Federal de São Paulo

Integrated Electronic Technician Program (High School Level) Jan 2017 – Dec 2020

IFSP - Instituto Federal de Educação, Ciência e Tecnologia de São Paulo

Experience

Software Developer April 2025– Present

Segura *Home Office*

- Development of Linux-based solutions for secure software communication
- Implementation of various communication and security protocols, including TCP, UDP, SSH, SFTP, HTTP/HTTPS, and SSL/TLS
- Hands-on experience with remote access and database protocols such as RDP (NLA), VNC, Telnet, and TDS
- Extensive use of large-scale open-source libraries and frameworks, including FreeRDP, Apache Guacamole, OpenSSH, and OpenSSL
- Architecting reliable and secure C code for mission-critical cybersecurity applications
- Active contributor to the open-source community, including bug fixes in the libssh2 library (mbedtls crypto backend) to enhance protocol stability and security

Systems Engineer Mar 2023– Mar 2025

AeroRiver *São José dos Campos*

- Development of Python, Bash scripts and C routines for reading and acquiring data from sensors
- Assembling electrical and electronic circuits
- Responsible for creating and organizing code versioning structure and agile development methodology
- Co-worker in the development of a Flight Aid System for WIG vehicle
- Configuration of Docker-based virtual development and testing environment
- Configuration of AWS EC2 and S3 instances for running StarCCM+ simulations
- Research Team Leader
- Implementation of unit test routines with GTEST

Programming and Robotics Instructor Nov 2024– June 2025

CtrlPlay *Taubaté*

- Game development with Scratch, C++, Python and JavaScript
- Planning and teaching programming lessons for children and teenagers
- Preparation of progress tracking reports

Freelance System Developer Jan 2021– Present

Freelancer *Multiple Locations*

- Development of a computer vision system for cattle detection and counting, assisting in automation of a farm management in Mato Grosso, Brazil
- Full-stack development of a web application for an online egg sales platform. Infrastructure setup, backend development, integration with payment platforms and database, frontend development
- Basic repairs and software installation on various operating systems

Mathematics Olympiad Project Instructor Jan 2019– Dec 2019

IFSP *São João da Boa Vista*

- Preparation and instruction of olympiad-related math classes
- Creation of assessment activities to test knowledge
- Mental preparation guidance for Mathematics Olympic

Profile

Systems Developer and Electronic Technician with a Bachelor's in Science and Technology from UNIFESP. Currently a Mid-Level Developer at Segura, specializing in high-performance Linux-based systems and network proxy implementation using C.

My background bridges the gap between hardware and software, with professional experience ranging from aerospace systems engineering at AeroRiver to developing complex middleware solutions. I have a proven track record in freelancing and technical instruction, demonstrating both technical versatility and the ability to communicate complex concepts effectively. I am also an active open-source contributor, with code merged into core libraries such as libssh2.

Technically proficient in C/C++, Bash, Lua, and Python, I focus on building efficient, low-level applications. My expertise in Linux internals, embedded systems, and electronics allows me to tackle challenges across the entire computing stack, from firmware to distributed systems.

Skills

C/C++	Docker
Python.....	Git.....
Lua	Arduino.....
Bash	2D&3D AutoCAD.....
Linux	

Projects

Distributed In-Memory Database - Full RESP Protocol Implementation | C++20

- Built a high-performance, multithreaded key-value store in C++20 that replicates core Redis functionalities. Implemented a custom TCP server with an event-driven architecture to handle concurrent connections using low-level socket programming.
- Developed a robust parser for the Redis Serialization Protocol (RESP) and integrated support for complex data structures, TTL-based key expiration, and RDB persistence for disk storage.
- Optimized memory usage and command execution latency, achieving performance metrics comparable to production-grade middleware systems.

SocialSTD - Rust-based automated Trello reporting project | CLI Tool

- Project developed in Rust that utilizes Trello API to request information about user's boards, lists and cards in order to generate a report featuring the most pertinent insights.

OrganizAI - Assistant & smart scheduler | Web Application

- OrganizAI is a project focused on use Artificial Intelligence to improve personal organization and time management. The project was a web application of an agenda, developed with Java as Back-end language and Vue.js framework as Front-end stack.

AI-Parking - Smart Parking management system using AI | Computer Vision

- Developed for the UNIFESP university campus, by utilizing computer vision techniques with convolutional networks using YOLOv8 model from Ultralytics, the project was developed to detect and identify vehicles in parking spaces. Using Python, the approach provides real-time monitoring and counting of available and occupied parking spots, offering a more efficient parking experience.

VPN-based Blockchain Network - Blockchain implementation using VPN approach | Blockchain

- Using a Hyperledger fabric blockchain and OpenVPN, this project aimed to compare time and space efficiency of implement a standard blockchain network versus one with VPN among the peers in order to assess the feasibility of adding an extra layer of security to blockchain network communication